

# Varsini Sakthivadivel Ramasamy

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## PROFESSIONAL SUMMARY

Bioinformatics Masters graduate specializing in AI/ML, Computational Biology, single-cell and spatial transcriptomics, with research interests in Alzheimer's disease and neurodegeneration. Experienced in building end-to-end scRNA-seq and multi-omics pipelines, applying deep learning and statistical models to high-dimensional biological data to extract mechanistic insight.

## EDUCATION

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| <b>Johns Hopkins University</b> – Baltimore, MD                                                                                            | Aug 2024 - May 2026 |
| Master of Science in Bioinformatics                                                                                                        | GPA: 3.89/4.0       |
| • <b>Coursework:</b> Genomic Data Science; Algorithms for Bioinformatics; Applied ML; Statistics; Gene Expression Analysis & Visualization |                     |
| <b>VIT University</b> – Vellore, India                                                                                                     | Sep 2020 - May 2024 |
| Bachelor of Technology in Biotechnology                                                                                                    | GPA: 3.96/4.0       |

## EXPERIENCE

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| <b>Research Assistant</b> – Dr. Maide Ozen, Johns Hopkins University School of Medicine                                                                                                                                                                                                                                                                                                      | Jan 2026 – Present   Baltimore, MD  |
| • Built an end-to-end pseudobulk RNA-seq pipeline to characterize microglial transcriptional heterogeneity across five developmental stages (E17–P60), integrating z-scored gene module scoring, PCA, differential gene expression, and pseudotime trajectory inference.                                                                                                                     |                                     |
| • Identified four distinct transcriptional programmes with coordinately shifting activities across microglial development, characterizing stage-specific regulatory transitions relevant to neuroinflammation and brain maturation.                                                                                                                                                          |                                     |
| <b>Research Assistant</b> – Johns Hopkins University School of Medicine                                                                                                                                                                                                                                                                                                                      | Dec 2025 – Apr 2026   Baltimore, MD |
| • Designed and trained a deep generative model (scVI; VAE) on 12,012 PBMC single cells to learn 30-dimensional latent embeddings for dimensionality reduction, UMAP visualization, Leiden clustering, and cell-type annotation of 12 PBMC populations.                                                                                                                                       |                                     |
| • Benchmarked scVI vs. PCA using controlled train/test splits, logistic regression, and a PyTorch MLP classifier; found PCA outperformed scVI on classification accuracy (LR:95.1% vs. 91.4%; 5-fold CV: 88.6% vs. 86.1%), while scVI produced biologically coherent latent structure, demonstrating that strong linear baselines can rival deep generative models on single-batch datasets. |                                     |
| <b>Research Assistant</b> – Johns Hopkins University                                                                                                                                                                                                                                                                                                                                         | Jun 2025 – Mar 2026   Baltimore, MD |
| • Developed an end-to-end ML pipeline for multiclass brain tumor classification using MRI radiomics features, training Logistic Regression, Random Forest, and XGBoost models with RandomizedSearchCV hyperparameter tuning and 5-fold cross-validation.                                                                                                                                     |                                     |
| • Applied SHAP to surface the most predictive radiomic features; XGBoost achieved 97.4% accuracy, Random Forest 95.5%.                                                                                                                                                                                                                                                                       |                                     |
| <b>Teaching Assistant</b> – Practical Computer Concepts, Johns Hopkins University                                                                                                                                                                                                                                                                                                            | Aug 2025 – May 2026   Baltimore, MD |
| • Supported instruction in UNIX, SQL, and Python for Practical Computer Concepts for Bioinformatics; developed automated RNA-seq and proteomics analysis pipelines in Python and R to aid course demonstrations and student workflows.                                                                                                                                                       |                                     |

## SKILLS

- **Machine Learning:** Supervised/unsupervised learning, Random Forests, representation learning, variational autoencoders (VAE), XGBoost, Deep learning (PyTorch), hyperparameter tuning, LLMs, model evaluation (ROC-AUC), feature engineering (SHAP)
- **Programming:** Python (pandas, scikit-learn, Biopython), R (tidyverse, Bioconductor), SQL, Bash, Linux/Unix, MatLab, AWS, Jupyter, Git.
- **Bioinformatics & Genomics Tools:** RNA-seq analysis, differential gene expression, single-cell RNA-seq (Scanpy, scVI), cell type annotation, transcriptomic data analysis, gene regulatory network analysis, NGS data analysis (Trimmomatic, Bowtie2/STAR), AutoDock, PyRx, GROMACS, Seurat, Discovery Studio, GEO, UCSC Genome browser, AlphaFold, MODELLER, Galaxy, Expasy, ProSA-web.
- **Data Science & Visualization:** Data pipelines, Statistical modeling, exploratory data analysis (EDA), multi-omics data analysis, high-throughput biological data analysis, pipeline development, gene regulatory network analysis, Matplotlib, Seaborn, ggplot2.

## PUBLICATIONS AND PROJECTS

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| <b>SpatialFormer: Spatially Aware Cell State Prediction</b> – Personal Project                                                                                                                                                                                                                                            | Apr 2026 – May 2026   JHU            |
| • Designed and trained SpatialFormer, a transformer jointly encoding each Visium spot with its 6 spatial neighbors ([CLS, query, nbr1–6]) via additive expression, coordinate, and token-type embeddings through a 2-layer, 4-head encoder                                                                                |                                      |
| • Achieved 89.1% validation accuracy on 10x mouse brain data (2,796 spots, 2,000 HVGs) and deployed an inference app on HuggingFace Spaces with attention and gene saliency interpretability. <a href="https://huggingface.co/spaces/Varsini-Sakthi/Spatialformer">huggingface.co/spaces/Varsini-Sakthi/Spatialformer</a> |                                      |
| <b>Genome Mining ML-Driven Screening of <i>Streptomyces</i> Metabolites</b> – Research Assistant                                                                                                                                                                                                                          | May 2023 – May 2024   VIT University |
| • Designed a computational discovery pipeline integrating genome mining, molecular docking, and machine learning scoring functions to identify antifungal metabolites from <i>Streptomyces</i> targeting scytalone dehydratase in <i>Magnaporthe oryzae</i> .                                                             |                                      |
| • Performed molecular docking and MD simulations using AutoDock, using GROMACS to evaluate protein–ligand stability ( <b>RMSD</b> , <b>RMSF</b> ) and applied similarity scoring and ML models ( <b>Rscoring</b> , <b>Kdeep</b> , <b>Gnina</b> ) identifying Cremomycin and Azetidomonamide B as inhibitors.              |                                      |